

Total No. of Questions : 10]

SEAT No. :

P3667

[5561]-300

[Total No. of Pages : 2

B.E. (I.T.)

INFORMATION AND CYBER SECURITY

(2012 Course) (End Semester) (414453)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

Q1) a) Find an integer of the following simultaneous equation using CRT. [6]

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 2 \pmod{7}$$

b) Explain active attack and passive attack. [4]

OR

Q2) a) Draw AES block diagram and state the general steps in detail. [6]

b) Distinguish between linear cryptanalysis and differential cryptanalysis. [4]

Q3) a) Enlist the requirement of hash function and explain working of MD-5 algorithm in detail. [6]

b) Describe the contents of Digital Certificate. [4]

OR

Q4) a) Explain Diffie Hellman key exchange algorithm with example. [6]

b) What four requirements were defined for Kerberos? [4]

Q5) a) Explain SSL hand shake and SSL Record protocol in detail with neat sketch. [8]

b) How AH and ESP differ while working under transport and tunnel mode. [8]

OR

P.T.O.

- Q6)** a) Explain different IDS methods with one example each. [8]
b) Define IKE protocol and explain its aggressive mode and quick mode in brief. [8]

- Q7)** a) What is cyber stalking? Explain the types of stalkers. [8]
b) Address security issues in cloud computing. [8]

OR

- Q8)** a) Describe the Indian and global legal perspective on cyber-crime. [8]
b) Explain with example how social engineering is playing wide role in cybercrime. [8]

- Q9)** a) What is SQL injection? Explain in detail. [6]
b) Write a short note on [12]
i) Different ways of password cracking.
ii) Indian IT Act 2000 and its challenges.

OR

Q10) Write notes on: [18]

- a) Phishing.
b) Host based malicious programs ; Trap Door, Logic Bombs and Trojan horse.
c) Key loggers & Spywares.

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